

3. Spread Locator

Q-1) what's statistical Distribution?

Ans) A statistical distribution describe how the values of a variable are spread or arranged and how frequently different values or ranges of value occur in probability, a distribution also tells us the probability associated with possible values of a random variable.

Q-2) what is a Q-Q plot and why is it used?

Ans) A Q-Q plot is a graphical method used to compare the distribution of a dataset with another distribution, often theoretical distributions such as the normal distribution.

- The main purpose is to check whether the data approximately follows a particular distribution, especially whether it is approximately normally distributed. it can also help identify differences in skewness, spread, tails and outliers.

Q-3) Difference between Discrete and continuous Distribution.

Discrete Distribution

Continuous Distribution

- | | |
|---|--|
| - value are countable | - Value can take any value within a range. |
| - represent counting | - represent measurement. |
| - Ex: Number of student passed. | - Ex: student's weight |
| - Possible values: 0, 1, 2, 3, ... | - Possible values: 60.1, 60.153, ... |
| - Uses a PMF | - Use a PDF |
| - probability can be assigned to an individual value. | - probability of one exact value is 0; probability is calculated over an interval/range. |

Q-4) what is Bernoulli Distribution?

Ans) The Bernoulli Distribution is a discrete probability Distribution used when a single experiment has only two possible outcomes, usually called success and failure.

- we represent the outcomes as:

$$X = \begin{cases} 1 & \text{Success} \\ 0 & \text{Failure} \end{cases}$$

Q-5) what is Binomial Distribution?

Ans) The Binomial Distribution is a discrete probability distribution used to find the probability of getting a certain number of successes in a fixed number of independent trials, where each trial has only 2 possible outcomes: success or failure.

- conditions:

- There is fixed number of Trials n .
- Each trial has two outcomes - success or failure.
- The trials are independent.
- The probability of success p remains the same for every trial.

Q-6) Explain Log- Distribution?

Ans) The log-normal distribution is a continuous probability distribution used for variable x when its logarithmic follows a normal distribution.

- The log-normal distribution is useful for variables that:
 - can only be positive ($x > 0$)
 - Are right-skewed
 - can have a few very large values.

- unlike the symmetric bell-curved normal distribution, a log-normal distribution typically has:
 - A peak toward the left
 - A long right tail
 - values only from 0 to infinity

Q-7) Explain Power Law Distribution

Ans) A power law distribution is a distribution where the probability of large values decrease according to a power of the value, rather than decreasing rapidly like in a normal or exponential distribution.

- power law distributions are useful when data contains rare but extremely large values and the usual bell shaped distribution does not describe the data well.

Q-8) what is Box-Cox Transform?

Ans) The Box-Cox transformation is a family of power transformation used to transform data, especially when the data is skewed or does not satisfy assumption such as normality. it can help make data more approximately normal and improve the usefulness of statistical tests and models.

Q-9) Explain Poisson distribution with an example.

Ans) The poisson distribution is a discrete probability distribution used to find the probability of a certain number of events occurring within a fixed interval of time or space, when events occurs at a known average rate and independently.

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The random variable represent count
 X = number of occurrence in an interval

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

x = number of occurrence,

λ = average number of occurrence in the interval.

Q-10) What is z-score probability?

Ans) A z-score tells us how many standard deviations a value is above or below the mean. Once we convert a value into z-score, we can use the standard normal distribution to find the probability associated with that value.

$$z = \frac{x - \mu}{\sigma}$$

x = observed value

μ = mean

σ = standard deviation

Q-11) Differentiate Probability Density function and Cumulative distribution

PDF	CDF
- Shows the density of probability around each value - written as $f(x)$ - probability is obtained from the area under the PDF - Used mainly to understand the shape of continuous data - PDF is not probability	- Shows the cumulative probability up to a value. - written as $F(x)$ - Directly gives $P(X \leq x)$ - used to find probability that X is less than or equal to value - CDF is probability.